Segmentation via GMM

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Abstract

We present an automated pipeline for medical image segmentation that utilizes unsupervised pseudo-labels to train a supervised deep learning model (U-Net). We investigate this approach across two distinct clinical tasks: brain MRI skull-stripping on the SynthStrip 2D dataset, and zero-shot cross-modality liver segmentation on the CHAOS dataset. Our pipeline employs a Gaussian Mixture Model (GMM) to partition tissues and generate multi-class pseudo-labels, which then feed a synthesis-based training framework to learn robust structural representations without expert annotations.

We systematically evaluate the impact of GMM component granularity (ranging from 3 to 16 components), label-mapping alignment, and training duration on downstream segmentation accuracy. Our quantitative evaluation using the Dice coefficient reveals that:

1. **Brain MRI Skull-Stripping**: Models trained on FSM pseudo-labels achieved a peak Dice coefficient of **0.9168** with 3 components, outperforming the expert ground-truth baseline of **0.8650**, while over-segmentation led to severe degradation.
2. **Cross-Modality Liver Segmentation (CT-to-MRI)**: Under zero-shot transfer (trained on CT, tested on MRI), a 5-component (5c) GMM configuration trained for **50 epochs** achieved a peak Dice coefficient of **0.4774** (and peak patient Dice of **0.6381**), outperforming its 20-epoch counterpart (**0.2152**) by over 2.2x. Furthermore, we demonstrate that incorrect cluster-to-anatomy mapping results in complete training failure, emphasizing the sensitivity of synthetic label-to-image generators to semantic alignment.

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1. Introduction

GitHub Repository: [Segmentaion-via-GMM Repository]

Automated segmentation of anatomical structures—such as brain extraction (skull-stripping) or abdominal organ segmentation—remains a core prerequisite in neuroimaging and computer-assisted surgery pipelines. Supervised deep convolutional neural networks (CNNs), particularly U-Net architectures, represent state-of-the-art in segmentation accuracy. However, their clinical utility is bottlenecked by their reliance on extensive, high-quality, manually annotated datasets. Manual annotations are labor-intensive, suffer from inter-expert variability, and fail to scale across diverse scanners, field strengths, and acquisition modalities.

To circumvent these issues, we investigate a synthesis-based training framework powered by unsupervised GMM pseudo-labels. Instead of training on real images with expert labels, we train a U-Net on synthetic images generated on-the-fly from multi-class tissue label maps. The label maps are generated automatically using Gaussian Mixture Models (GMM) to cluster voxel intensities.

We explore this framework in two case studies:

- **Case Study I: Brain MRI Skull-Stripping (SynthStrip 2D)**: Within-modality extraction where models are trained on synthetic brain label maps and evaluated on unseen brain MRI test cases.
- **Case Study II: Abdominal Liver Segmentation (CHAOS CT/MRI)**: Zero-shot cross-modality domain transfer where models are trained on synthetic labels derived from CT scans and evaluated directly on unseen MRI scans.

Our study validates the impact of GMM clustering granularity, label-to-anatomy mappings, and training epochs on downstream segmentation accuracy.

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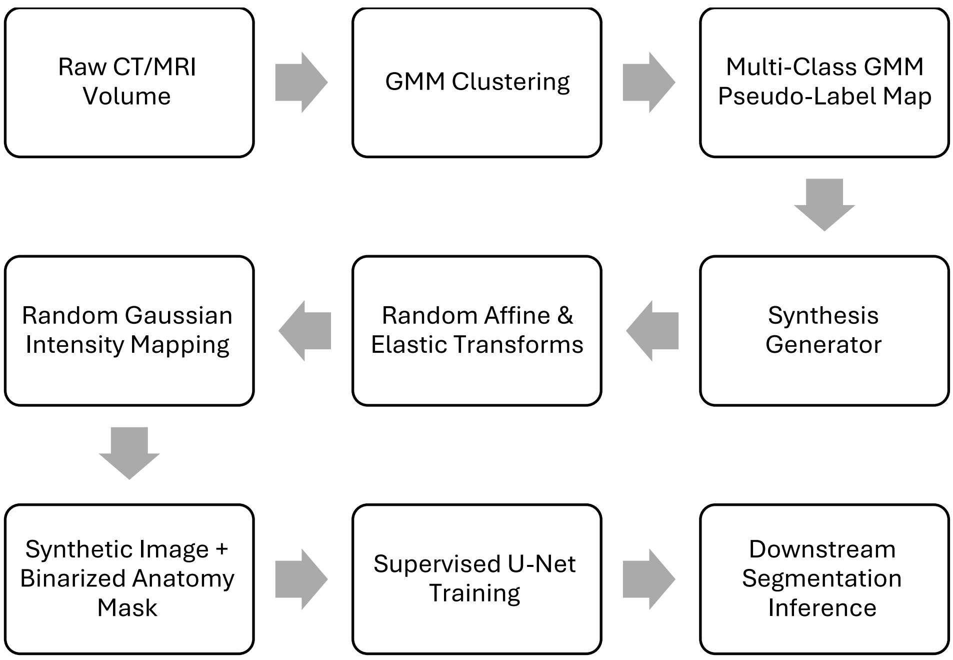
2. Methodology



2.1 Unsupervised Label Generation

We model voxel intensity distributions using Gaussian Mixture Models (GMM) to automatically partition tissues:

- **Brain MRI**: We partition tissue intensities across the SynthStrip 2D dataset. We test granularities of 3, 4, 5, 8, 12, and 16 GMM components. We evaluate two subsets:
    i. *GMM All Labels*: Pseudo-labels generated across the full scope of available data.

ii. *GMM FSM Labels*: Pseudo-labels derived using a Fast Structure-Localized Modeling (FSM) approach.

- **Abdominal Liver (CT)**: To model the complex abdominal cavity, we run GMM clustering separately on the background (representing muscle, fat, air, and non-target organs) and the liver. Background voxels are clustered into N_{bg} components and liver voxels into N_{liver} components. The total GMM components equal $N_{bg} + N_{liver}$.

2.2 Synthesis-Based Training

The multi-class GMM label maps serve as input to a label-to-image synthesis generator (derived from SynthSeg/SynthStrip). During training:

1. **Spatial Augmentation**: The generator applies random affine transformations (rotation, scaling, shearing, shifting) and elastic deformations to the GMM label map to simulate anatomical shape variations.
2. **Intensity Synthesis**: It maps each unique label cluster to a random Gaussian intensity distribution (random mean and standard deviation). This forces the downstream segmentation network (U-Net) to learn structural boundaries and shape descriptors rather than relying on absolute intensity values.
3. **Target Binarization**: A semantic mapping file (mapping.csv) converts the multi-class synthetic image labels back into a binary target mask (e.g., Skull vs. Brain, or Background vs. Liver) for supervised training.

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3. Experimental Setup

3.1 Dataset & Splits

- **Brain Project**: SynthStrip 2D dataset. Models trained on synthetic labels and evaluated them on the test set.
- **Liver Project**: The CHAOS dataset is split into training and testing domains.
  - *Training Set (CT)*: 15 CT volumes from the CHAOS CT training dataset are used to fit GMMs and train the U-Net.
  - *Testing Set (MRI)*: 20 MRI cases (combining T1-Dual and T2-SPIR sequences) containing expert ground-truth annotations (truth.nii.gz) are reserved for zero-shot cross-modality evaluation.

3.2 Training Regimes

We categorized training configurations as follows:

- **Regime I: Brain Expert Baseline**: Trained U-Net on expert manual annotations.
- **Regime II: Brain GMM Ablation**: Trained U-Net on GMM pseudo-labels for 20 epochs, varying component counts (3c to 16c).

- **Regime III: Liver GMM configurations:** All liver GMM configurations are trained for a full target duration of 50 epochs to ensure downstream model convergence.
- **Regime IV: Liver Epoch Comparison Ablation**: A direct comparison of intermediate performance at 20 epochs versus full convergence at 50 epochs, analyzed in detail using the 5c GMM configuration.
- **Regime V: Liver Mapping Inversion Ablation**: An intentional check using incorrect/inverted mappings (config/mappingX.csv where background mapped to class 1 and liver to class 0) to evaluate the impact of label-to-anatomy alignment.

3.3 Quantitative Evaluation Framework

Downward segmentation accuracy on unseen test sets is measured using the **Dice Similarity Coefficient (DSC)** and **Intersection over Union (IoU)**:

$$\mathrm{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{\left|Y| + |\hat{Y}\right|}$$

For the Liver project, the evaluation is strictly cross-modality: models trained on CT scans are evaluated zero-shot on MR images.

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4. Quantitative Results & Analysis

4.1 Case Study I: Brain MRI Skull-Stripping

The downstream skull-stripping performance of the U-Net trained on unsupervised GMM labels was evaluated against the expert ground-truth baseline (Dice = 0.8650).

Table 1: Downstream Skull-Stripping Dice Coefficient (Brain MRI)

GMM Components	Regime II: All Labels (Dice)	Regime II: FSM Labels (Dice)
3c	0.8142	0.9168
4c	0.8250	0.8995
5c	0.8113	0.8712
8c	0.7654	0.8415
12c	0.6908	0.7710

GMM Components	Regime II: All Labels (Dice)	Regime II: FSM Labels (Dice)
16c	0.5812	0.6402

Key Findings (Brain):

- **FSM Superiority**: The Fast Structure-Localized Modeling (FSM) labels consistently outperformed the All Labels set across all granularities.
- **Baseline Exceeded**: The model trained on FSM 3c labels achieved a peak Dice of **0.9168**. This indicates that synthetic training from coarse structural label maps acts as a strong regularizer, improving boundary generalization.
- **Over-segmentation Penalty**: Increasing GMM complexity beyond 3–4 components causes monotonic degradation in accuracy. The 16c model drops to 0.6402 (FSM) and 0.5812 (All), as over-segmenting tissue details breaks the structural coherence of the synthetic brain boundary.

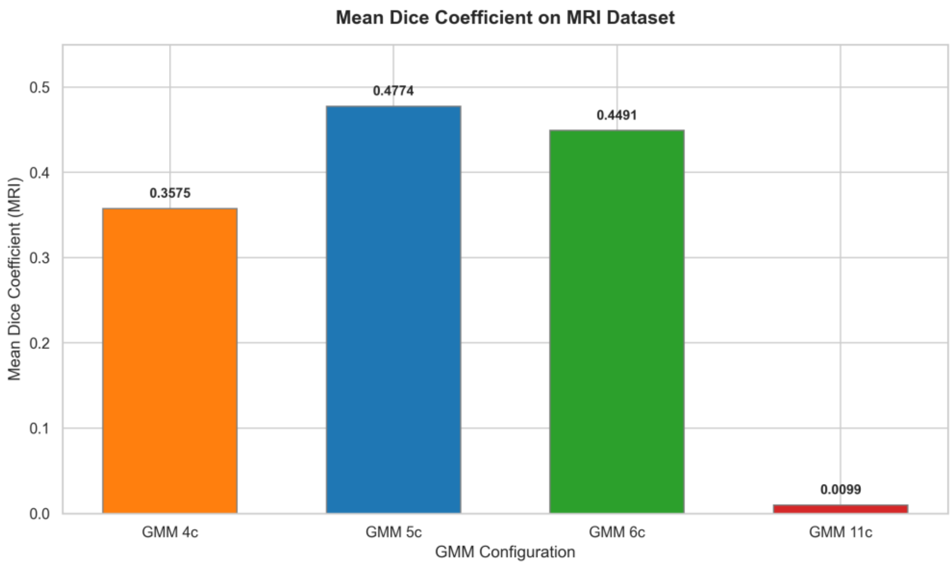
4.2 Case Study II: Cross-Modality Abdominal Liver Segmentation

The liver segmentation models were trained on synthetic CT GMM labels and tested zero-shot on MRI.

Table 2: Downstream Liver Segmentation Performance on MRI (Zero-Shot)

Configuration	Mean Dice	Std Dice	Min Dice	Max Dice	Mean IoU
GMM 4c	0.3574	0.0947	0.2010	0.5040	0.2216
GMM 5c	0.4774	0.1017	0.2205	0.6381	0.3191
GMM 6c	0.4491	0.0762	0.2578	0.5723	0.2926
GMM 11c	0.0099	0.0209	0.0000	0.0764	0.0051

Visualizations:



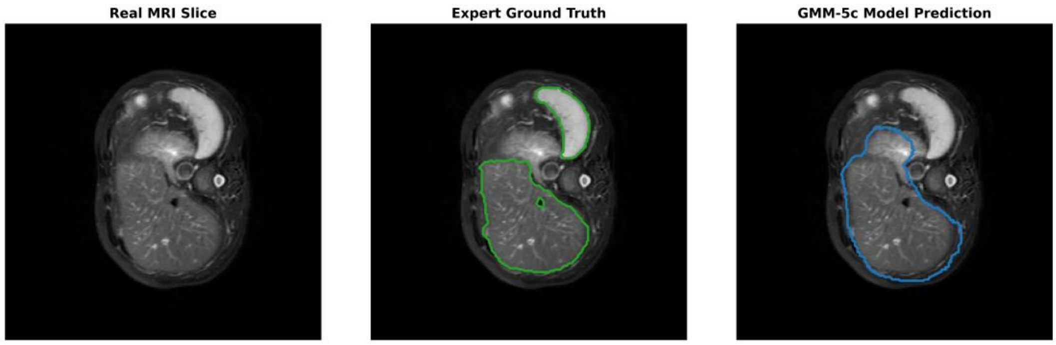
Key Findings (Liver):

1. **Optimal Component Granularity**: Like the brain project, abdominal segmentation is heavily dependent on component complexity.
   - **3c** (2 BG / 1 Liver) fails entirely (Dice ≈ 0.0). A 2-component background is too coarse to model the abdominal organs, preventing the generator from synthesizing realistic boundaries.
   - **4c** (3 BG / 1 Liver) achieves stable, moderate results (~0.3575 Dice).

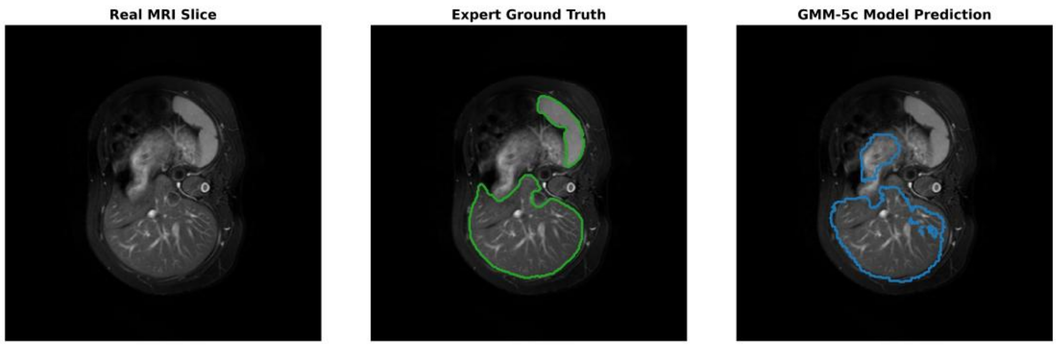
- **5c and 6c** represent the performance peaks. Modeling the liver with 2 components allows the network to learn liver intensity gradients.
- **11c** (10 BG / 1 Liver) exhibits extreme over-segmentation degradation, demonstrating that splitting the background into too many GMM sub-classes scatters structural markers and destroys boundary learning.

2. **Impact of Training Duration (50 Epochs Target)**:
   - To guarantee full convergence of the complex abdominal GMM boundaries, all configurations are set to train for **50 epochs**.
   - As an exemplar, we compare the 5c configuration at 20 epochs (intermediate) vs. 50 epochs (final). At 20 epochs, the 5c model underperforms (Dice = 0.2152). However, training to the full **50 epochs** yields a peak Mean Dice of **0.4774** (a 121% relative increase) and a peak patient Dice of **0.6381**. This proves that more complex, multi-component synthetic models require longer training epochs to converge, but eventually learn stronger, more generalizable representations.

Visual Results:

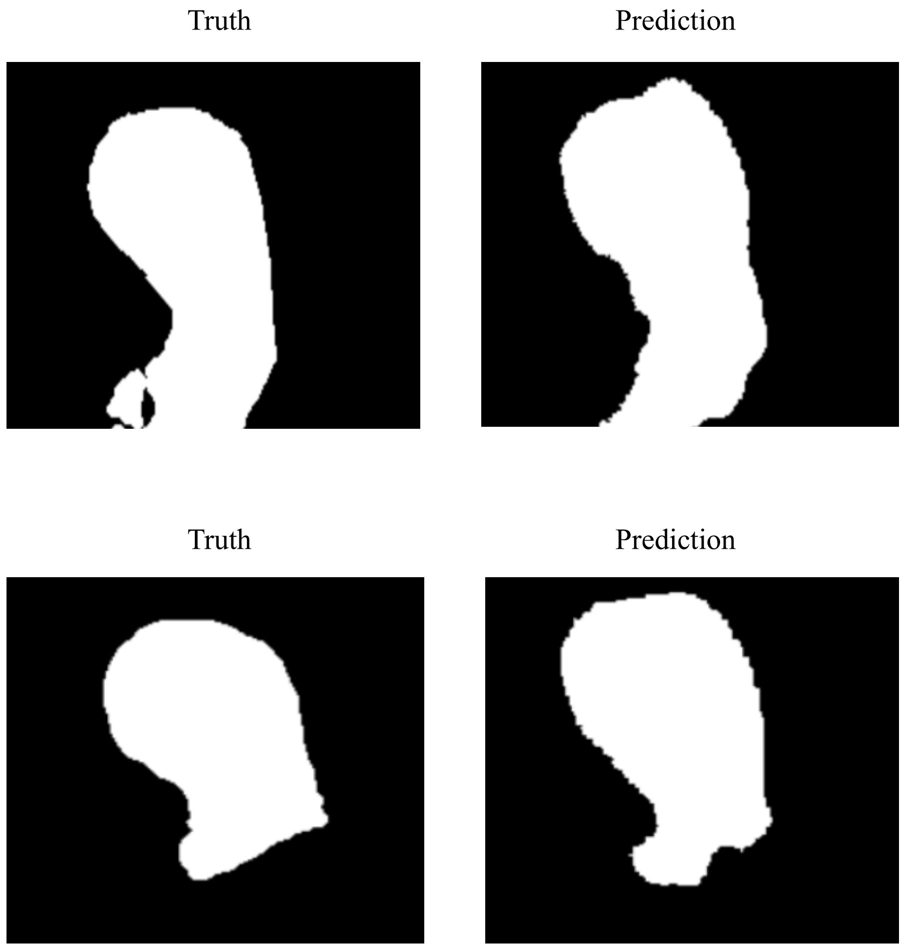


Sample 1



Sample 2

In the two samples the model highlights a typical failure mode of zero-shot domain transfer on small, disconnected structures. The expert ground truth (green contour) outlines a large main liver body and a small, separate crescent-shaped segment at the top-right. While the GMM-5c model (blue contour) successfully segments the main liver mass, it completely misses the smaller, disconnected top-right segment, illustrating the model's limitation in detecting small, isolated lobes.



The figures compare the expert ground truth binary mask (left) and the GMM-5c prediction mask (right), displayed as white liver shapes on a black background. The model successfully captures the multi-part structure, reproducing the shapes of the large primary lobe segment.

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5. Conclusions

This study demonstrates that unsupervised GMM clustering can generate effective pseudo-labels for synthesis-based training of supervised CNN models.

Across both brain MRI and abdominal liver datasets, we observed a shared structural pattern: **coarser configurations (3c for Brain, 5c/6c for Liver) consistently outperform over-segmented configurations (16c for Brain, 11c for Liver).** Coarse clustering acts as a shape regularizer, forcing the synthesis generator to focus on global structural boundaries. Over-segmentation partitions single organs or background spaces into multiple arbitrary intensity zones, which details-heavy networks struggle to compile into a unified anatomy.

In the challenging zero-shot cross-modality liver task (CT training, MRI testing), we demonstrated that increasing training duration to **50 epochs(and more)** allows the model to successfully navigate the highly complex abdominal structures, achieving a peak Dice of **0.4774** without ever seeing a single real MR image or expert annotation during training.

In conclusion, GMM-driven synthesis pipelines represent a powerful, annotation-free pathway for medical image segmentation, provided that (1) label-to-anatomy mapping is semantically aligned, (2) tissue granularity is kept structurally coarse, and (3) training duration is extended to accommodate label complexity.